NEWS AND SOCIAL MEDIA ANALYTICS ENGINE

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



DECLARATION

The Mini Project Report entitled "News and Social Media Analytics Engine**"** is a record of Bonafide work of **J Pragathi – 2520090216, J Rakshitha – 2520090212,** submitted in partial fulfillment for the award of B. Tech in Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the mini project based report entitled "**News and Social Media Analytics Engine"** is a bonafide work done and submitted by **J Pragathi – 2520090216, J Rakshitha – 2520090212** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Computer Science and Information Technology, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

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J Pragathi – 2520090216

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1.ABSTRACT

News and social media platforms contain large amounts of information that can be difficult to search, organize, and review manually. The **News and Social Media Analytics Engine** is a web-based prototype designed to demonstrate how data structures and algorithms can help organize and analyze news and social media posts. The system provides features such as searching posts, detecting duplicate content, ranking posts according to priority, grouping related stories, managing posts for moderation, and displaying relationships between posts or topics.

The backend is developed using Java, while HTML, CSS, and JavaScript are used to create the frontend. Node.js with Express is used to connect the web interface with the Java processing component. A CSV file provides the sample dataset used by the application. The project applies hash tables for key-based searching and duplicate detection, heaps for priority ranking, queues for managing items awaiting review, trees for organizing topic information, and graphs for representing relationships. A clustering module groups posts according to the implemented logic.

The project combines these concepts in a single dashboard, demonstrating the practical application of data structures and algorithms in content organization. The current implementation uses sample data and serves as an educational prototype. Future improvements may include live data integration, advanced text similarity techniques, database storage, and support for larger datasets.

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2. INTRODUCTION

2.1 Background

News websites and social media platforms generate a large amount of text-based information every day. Users often face difficulties in finding specific posts, identifying repeated content, grouping related stories, and deciding which information requires attention first. Organizing such information manually can be time-consuming.

Data Structures and Algorithms provide efficient methods for storing, searching, organizing, and processing information. By combining different data structures, applications can perform specific operations in a more organized manner. This project applies these concepts to a news and social media content-processing scenario.

2.2 Project Overview

The News and Social Media Analytics Engine is a web-based application that processes a sample collection of posts and displays the results through an interactive dashboard. The application includes different sections for viewing posts, searching content, detecting duplicates, ranking posts, grouping related stories, managing moderation, and viewing relationships.

The system uses Java to implement the data-processing modules. HTML, CSS, and JavaScript provide the user interface, while Node.js and Express support communication between the frontend and backend. The sample posts are supplied through a CSV file.

2.3 Motivation

The main motivation behind this project is to apply the concepts learned in Data Structures and Algorithms to a practical application. Instead of implementing data structures separately, the project combines them to demonstrate how they can work together in a content-processing system.

2.4 Scope of the Project

The current project focuses on processing a sample dataset and demonstrating different operations through a local web dashboard. Its scope includes searching, duplicate detection, priority ranking, story grouping, moderation-related processing, and relationship visualization.

The current version is a prototype and does not automatically collect live news or social media posts from external platforms.

2.5 Organization of the Report

This report describes the literature survey, problem statement, objectives, hardware and software requirements, implementation of the project modules, results obtained from the working application, and the conclusion and future scope of the project.

3. LITERATURE SURVEY

3.1 Google News

Google News is a news aggregation platform that allows users to explore articles from different publishers. It organizes news coverage around topics and current events, helping users discover information from multiple sources. Its approach provides a useful reference for understanding news discovery and the organization of related stories. Our project demonstrates similar organizational ideas on a smaller scale using a sample dataset. It does not reproduce the complete aggregation or ranking system used by Google News.

3.2 GDELT Project

The GDELT Project collects and organizes information about news coverage from different parts of the world. It supports research into events, topics, entities, and trends reported in news media. The project demonstrates how large collections of news information can be processed for analysis. This provides background for building systems that organize news content. Our implementation currently uses a local CSV dataset rather than live GDELT data.

3.3 NewsAPI

NewsAPI provides interfaces for searching news articles and retrieving headlines from supported sources. Its documentation describes how applications can request articles and apply search criteria. This is relevant to the search and content-retrieval aspects of our project. NewsAPI could be integrated in a future version to obtain current news articles. The current prototype uses sample data and does not claim to have a live NewsAPI integration.

3.4 Reddit API

The Reddit API provides programmatic access to Reddit content through supported endpoints and platform rules. Developers can use its documentation to understand how posts and comments may be retrieved by an application. Such data access can support future social media analysis systems. This resource is relevant to the possible extension of our project to external social media data. Our present implementation demonstrates content processing using a local dataset instead.

3.5 Java Collections Framework

The Java Collections Framework provides interfaces and classes for representing groups of objects and performing operations on them. It includes structures such as lists, queues, sets, and priority queues. These concepts are relevant to applications that need to store, retrieve, and organize records. Studying the framework helps explain how data structures can support application workflows. Our project uses Java modules to demonstrate these ideas in a content-processing context.

3.6 Java Priority Queue

Java's PriorityQueue documentation explains the operations and behavior of a priority-based collection. Priority queues are useful when items must be processed according to their priority rather than only their arrival order. This concept is relevant to the priority-ranking feature of our project. A heap can efficiently maintain the highest-priority item for retrieval. Our project includes a max-heap module intended to demonstrate priority-based ordering.

3.7 HTML

HTML is the standard markup language used to structure web pages. It defines page elements such as headings, sections, links, forms, and content containers. These elements provide the basic structure of a web interface. In our project, HTML is used to create the landing page and dashboard layout. It works with CSS and JavaScript to provide the complete frontend experience.

3.8 CSS

CSS is used to control the appearance and layout of web pages. It allows developers to define colors, typography, spacing, alignment, and other visual properties. Proper styling improves the readability and usability of a dashboard. In our project, CSS is used to style the landing page and the different dashboard sections. It helps present the application's features in a consistent interface.

3.9 JavaScript

JavaScript enables interactive behavior in web applications. It can respond to user actions, update page content, and communicate with backend services. It is commonly used alongside HTML and CSS to build interactive dashboards. In our project, JavaScript supports frontend interactions and requests to the server. This allows the user to navigate the application and view the results of processing operations.

3.10 Express.js

Express.js is a web application framework for Node.js that supports HTTP routes and request handling. It is commonly used to build backend services and connect web interfaces with application logic. Routes can receive requests and return data to the frontend. In our project, Node.js with Express acts as the integration layer between the browser interface and the Java processing component. It helps connect the dashboard with the project's processing features.

4. Problem Statement and Objectives

Problem Statement:

Organizing, searching, and reviewing news and social media posts becomes difficult when repeated or related content is not systematically identified, grouped, and prioritized.

Problem Description:

Users need a convenient way to search posts, detect duplicate content, identify high-priority items, group related stories, and manage content for review. The project demonstrates how data structures and algorithms can support these tasks using a sample dataset.

Objectives:

1. To develop a web-based dashboard for displaying and organizing sample news and social media posts.
2. To implement searching and hash-based duplicate detection.
3. To use a max-heap-based module for priority ranking.
4. To organize topic information using a tree structure.
5. To group related posts using the implemented clustering logic.
6. To represent moderation work through a queue-based workflow.
7. To represent relationships between posts or topics using a graph.
8. To connect the frontend with the Java processing component using Node.js and Express.
9. To demonstrate the practical application of Data Structures and Algorithms in a working prototype.

Expected Outcome

The expected outcome is a working local application that provides an interactive dashboard for exploring sample posts and demonstrating the implemented search, duplicate detection, priority ranking, story grouping, moderation, and relationship features.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Processor: Intel Core i3 or equivalent and above.
- RAM: Minimum 4 GB; 8 GB recommended.
- Storage: At least 10 GB of available storage.
- Display: Standard laptop or desktop display.
- Network: Internet connection for installing development tools and accessing documentation.

Software Requirements:

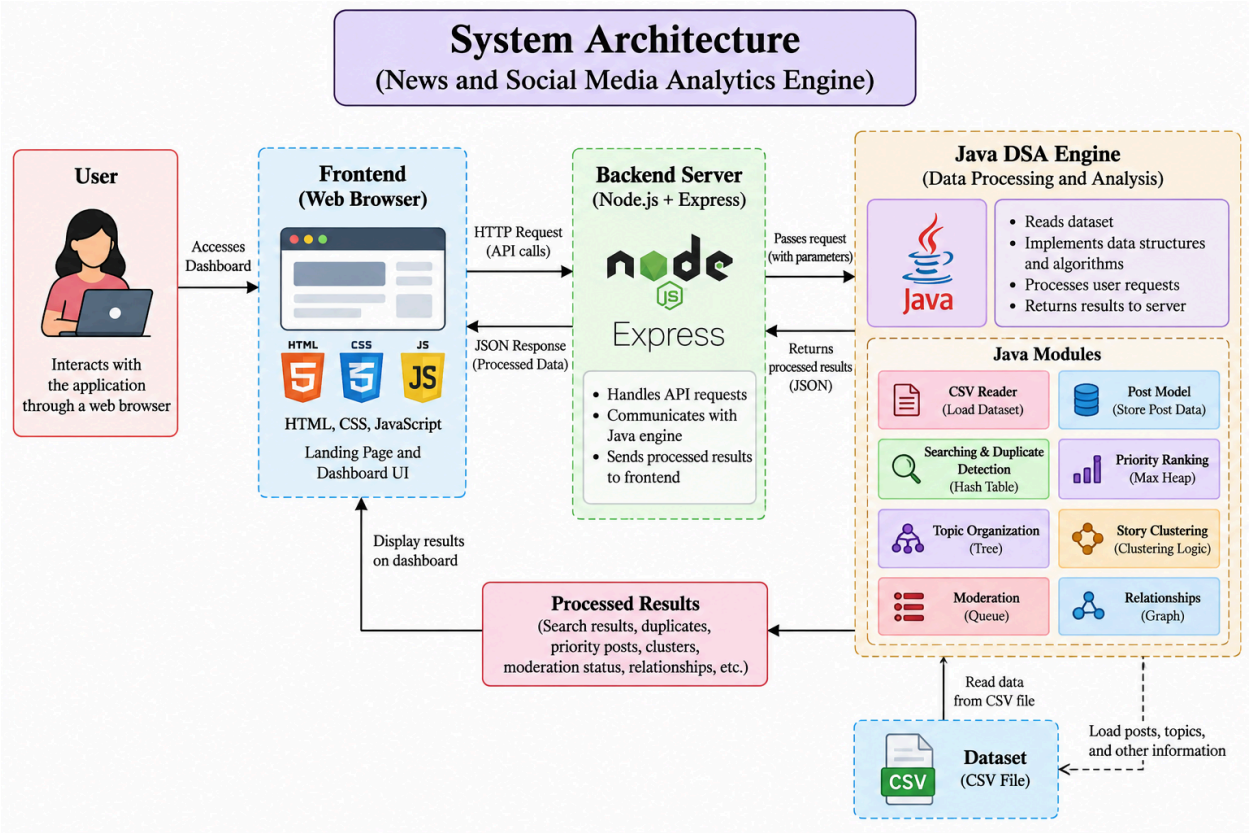
- Operating System: Windows or another operating system that supports Java and Node.js.
- Programming Language: Java.
- Frontend Technologies: HTML, CSS, and JavaScript.
- Backend Integration: Node.js and Express.js.
- Data Input: CSV file containing sample posts.
- Code Editor: Visual Studio Code or another compatible development environment.
- Version Control: Git and GitHub, if used.

System Architecture

The application follows a frontend-backend architecture. The user interacts with the dashboard through a web browser. The frontend communicates with the Node.js and Express server, which connects to the Java processing component. The Java engine processes the sample data using the implemented modules. The resulting output is then displayed in the dashboard.

User → Web Browser → HTML/CSS/JavaScript Frontend → Node.js/Express Server → Java DSA Engine → Processed Results → Dashboard Display

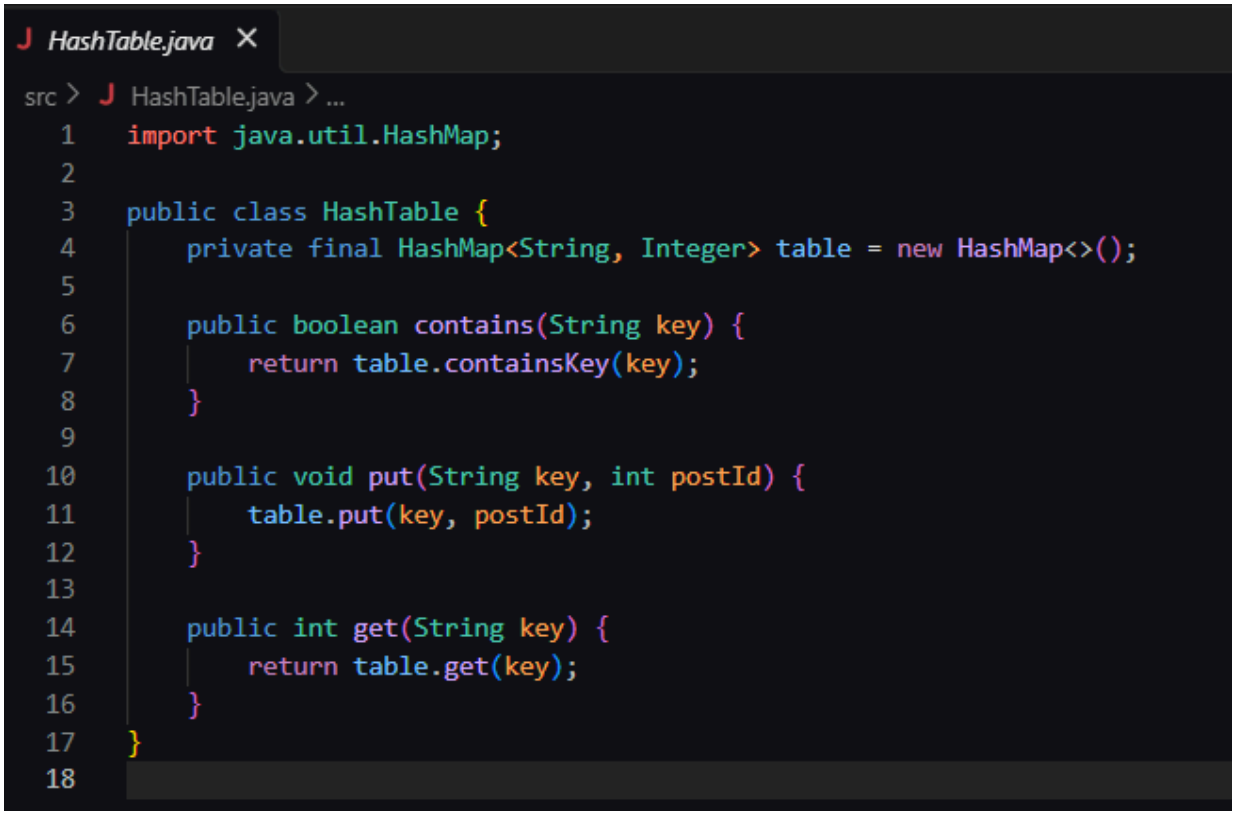
The Java engine includes modules for reading CSV data, representing posts, searching and duplicate detection, priority ranking, topic organization, clustering, moderation, and relationships.



6. IMPLEMENTATION

HashTable.java — Searching and Duplicate Detection

Include the methods that insert keys, search for a key, and check for duplicate posts. Add a short explanation of how hashing supports efficient lookup.



MaxHeap.java — Priority Ranking

Include the methods for inserting posts into the heap and retrieving/removing the highest-priority post. Explain how the heap places higher-priority posts first.

MaxHeap.java

```java
import java.util.ArrayList;

public class MaxHeap {
    private final ArrayList<Post> heap = new ArrayList<>();

    public void insert(Post post) {
        heap.add(post);
        int i = heap.size() - 1;

        while (i > 0) {
            int parent = (i - 1) / 2;
            if (heap.get(parent).priority >= heap.get(i).priority) break;
            swap(parent, i);
            i = parent;
        }
    }

    public Post removeMax() {
        if (heap.isEmpty()) return null;

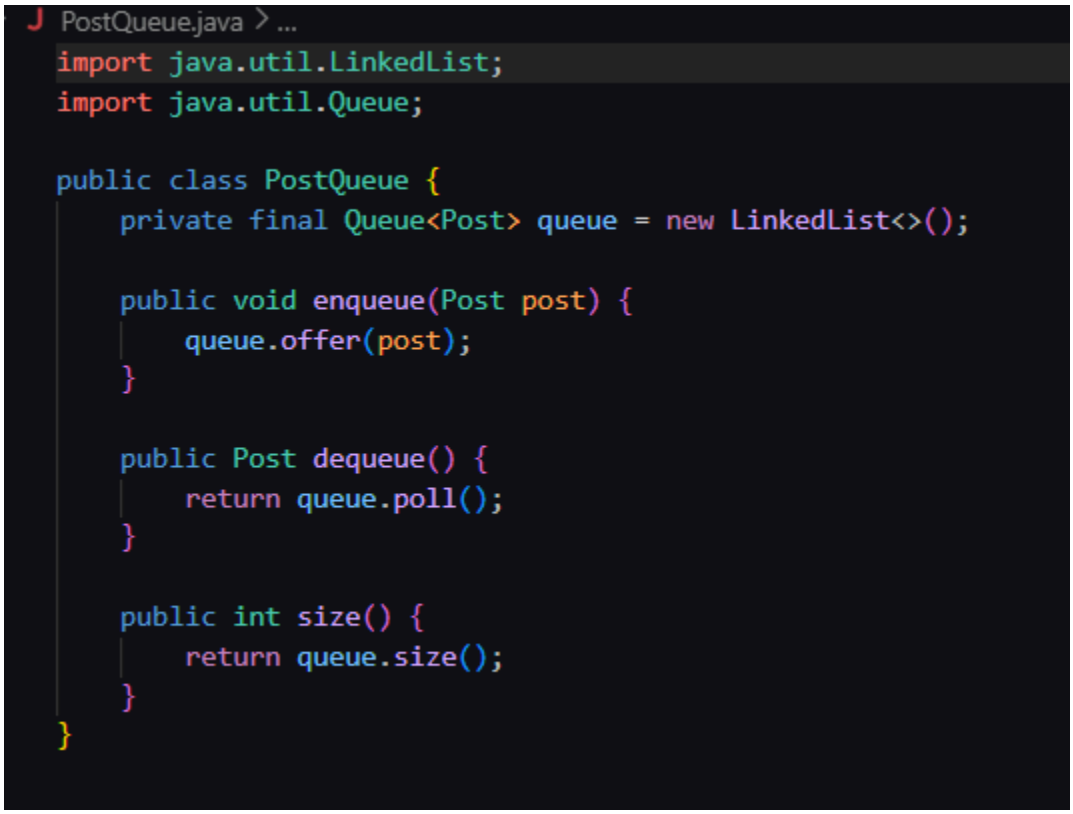
        Post max = heap.get(index: index: 0);
        Post last = heap.remove(heap.size() - 1);

        if (!heap.isEmpty()) {
            heap.set(index: index: 0, last);
            heapify(i: i: 0);
        }
        return max;
    }
```

```
31      private void heapify(int i) {
32          while (true) {
33              int left = 2 * i + 1;
34              int right = 2 * i + 2;
35              int largest = i;
36
37              if (left < heap.size() &&
38                  heap.get(left).priority > heap.get(largest).priority)
39                  largest = left;
40
41              if (right < heap.size() &&
42                  heap.get(right).priority > heap.get(largest).priority)
43                  largest = right;
44
45              if (largest == i) break;
46              swap(i, largest);
47              i = largest;
48          }
49      }
50
51      private void swap(int a, int b) {
52          Post temp = heap.get(a);
53          heap.set(a, heap.get(b));
54          heap.set(b, temp);
55      }
56  }
57
```

3. PostQueue.java — Moderation Queue

Include the enqueue and dequeue methods, if present. Explain how a queue manages posts waiting for moderation.



4. Graph.java — Relationships

Include the methods that create vertices, add edges, or retrieve relationships. Explain how the graph represents connections between posts or topics.

Graph.java > ...

```java
import java.util.*;

public class Graph {
    private final Map<String, Set<String>> adjacency = new HashMap<>();

    public void addEdge(String a, String b) {
        adjacency.computeIfAbsent(a, k -> new HashSet<>()).add(b);
        adjacency.computeIfAbsent(b, k -> new HashSet<>()).add(a);
    }

    public void display() {
        System.out.println(x: x: "\n--- RELATIONSHIP GRAPH ---");
        for (String node : adjacency.keySet())
            System.out.println(node + " -> " + adjacency.get(node));
    }
}
```

7. RESULTS & DISCUSSION

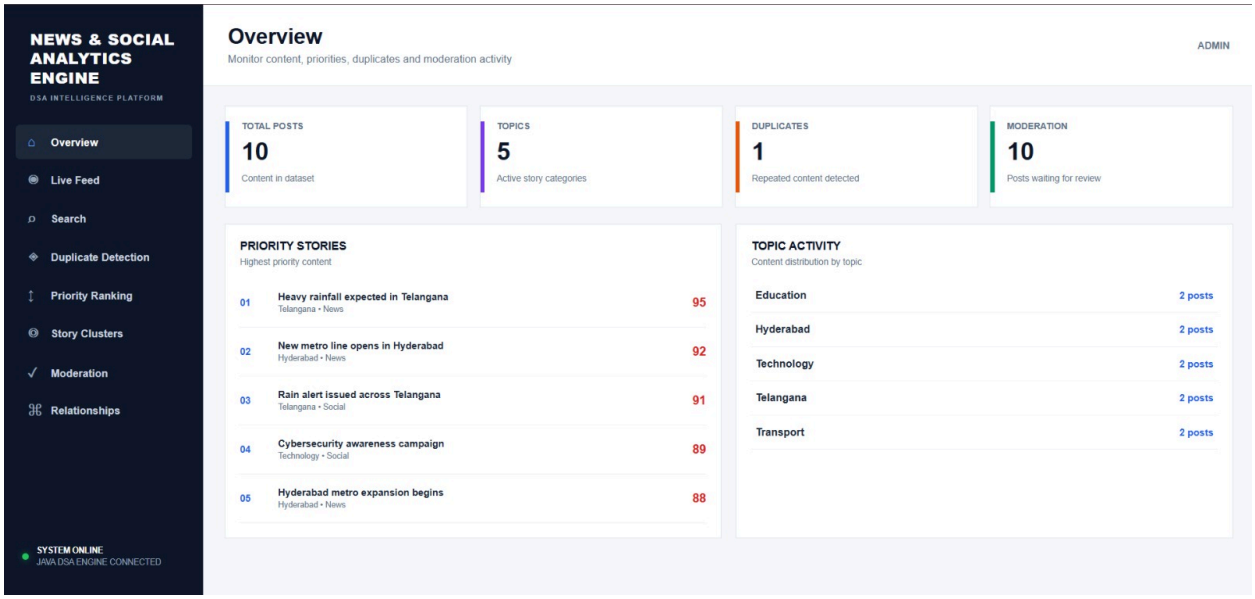
Landing Page

The landing page introduces the News and Social Media Analytics Engine and provides a "LET'S START" button to access the dashboard. It presents the purpose of the application through a clean and user-friendly interface.



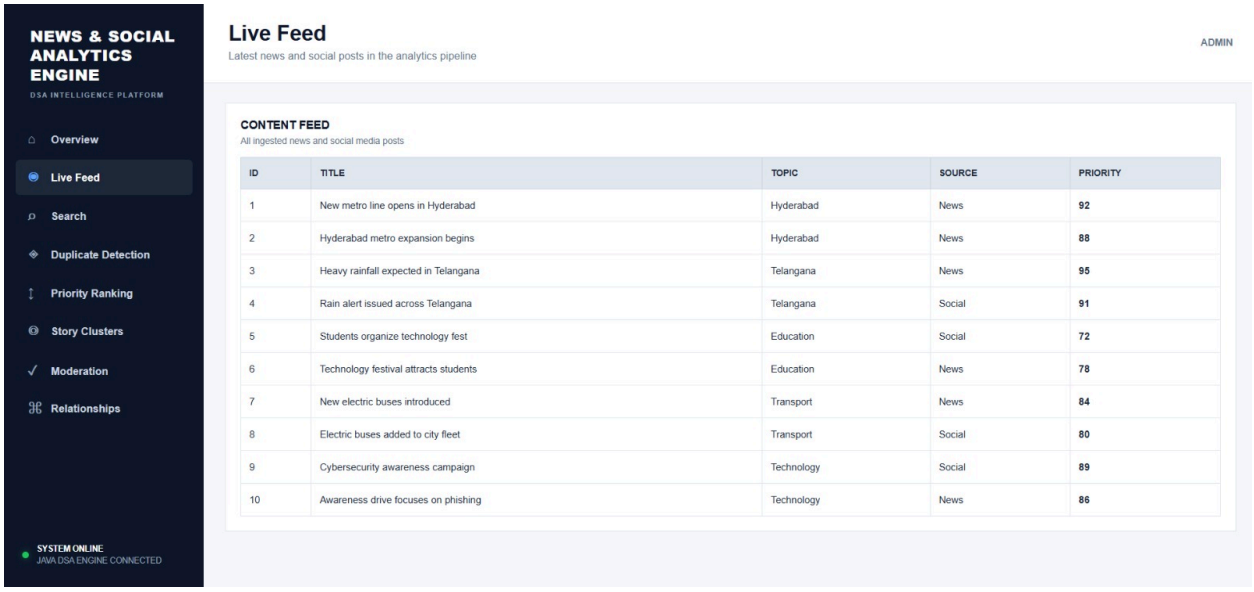
Overview Dashboard

The Overview dashboard displays summary information such as total posts, topics, duplicates, and moderation activity. It also highlights priority stories and shows the distribution of posts across topics.



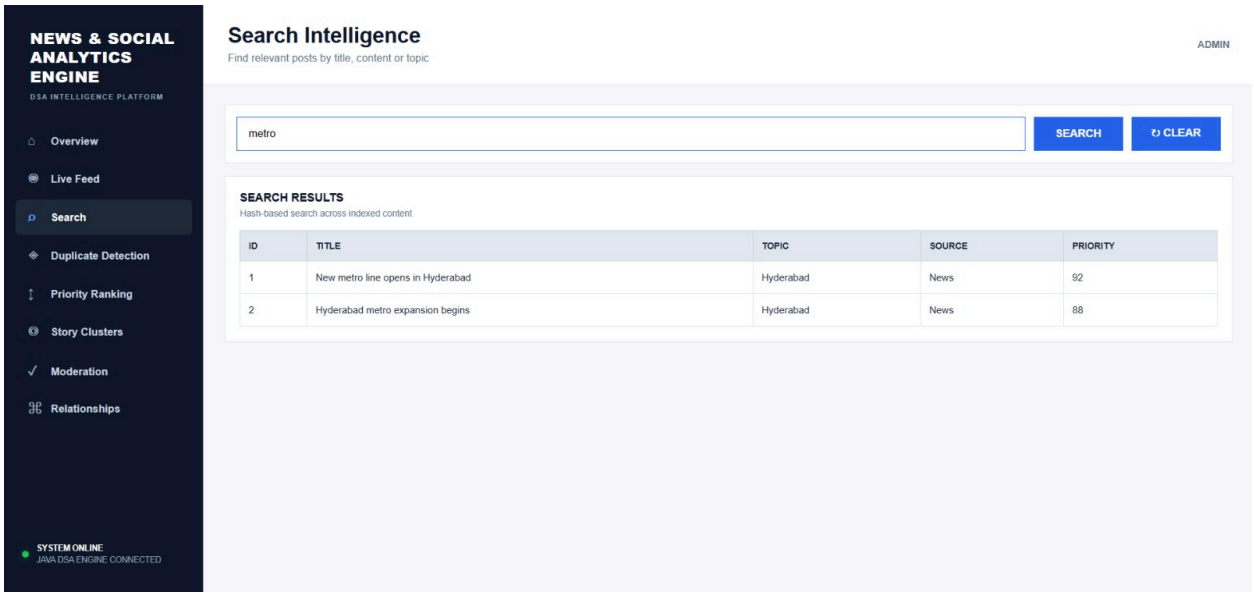
Live Feed

The Live Feed displays the available news and social media posts along with their IDs, titles, topics, sources, and priority scores. It allows users to view and understand the sample dataset in an organized format.



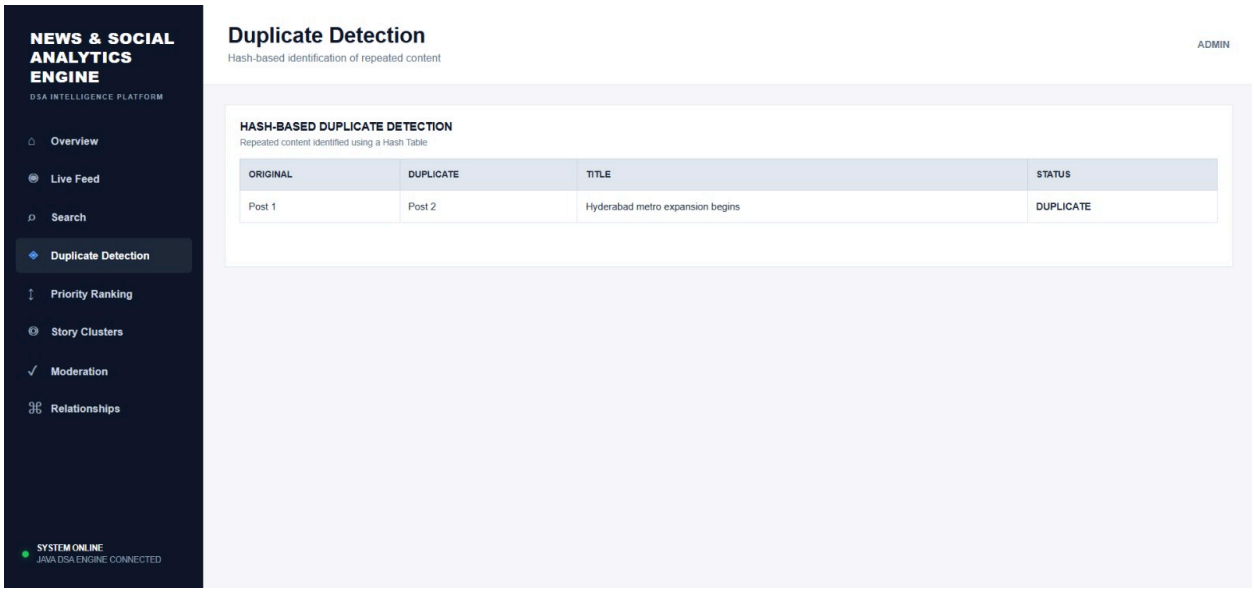
Search

The Search section allows users to find relevant posts by entering a keyword. In this example, searching for “metro” returns matching posts from the dataset.



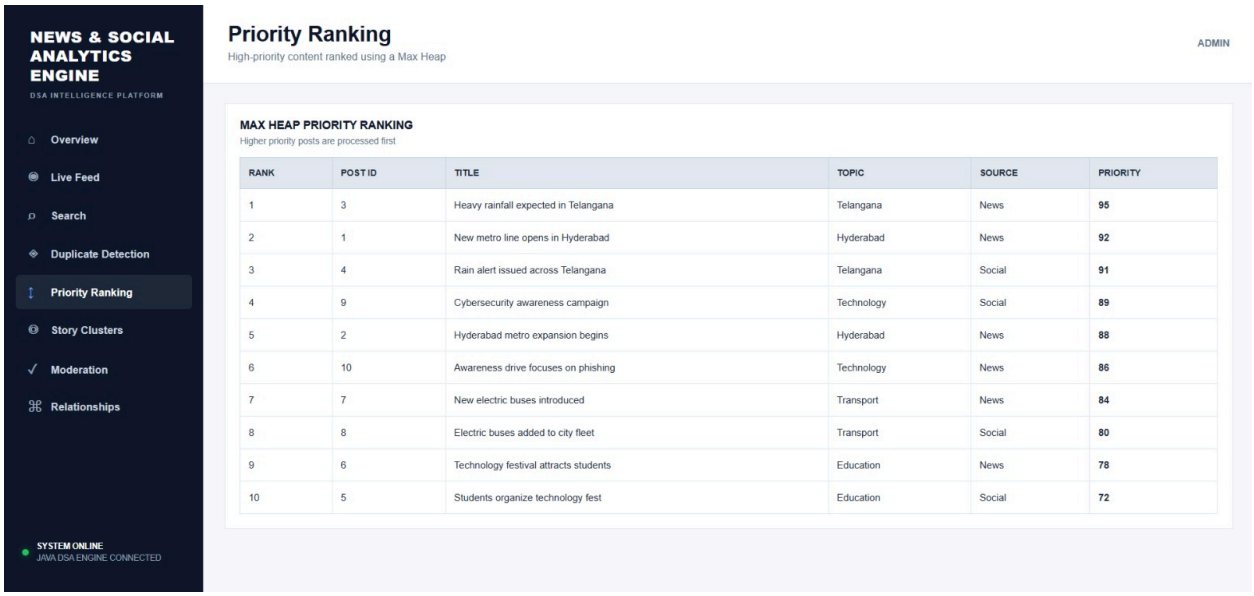
Duplicate Detection

The Duplicate Detection module identifies repeated content using hash-based checking. The results display the original post, its duplicate, and the corresponding duplicate status.



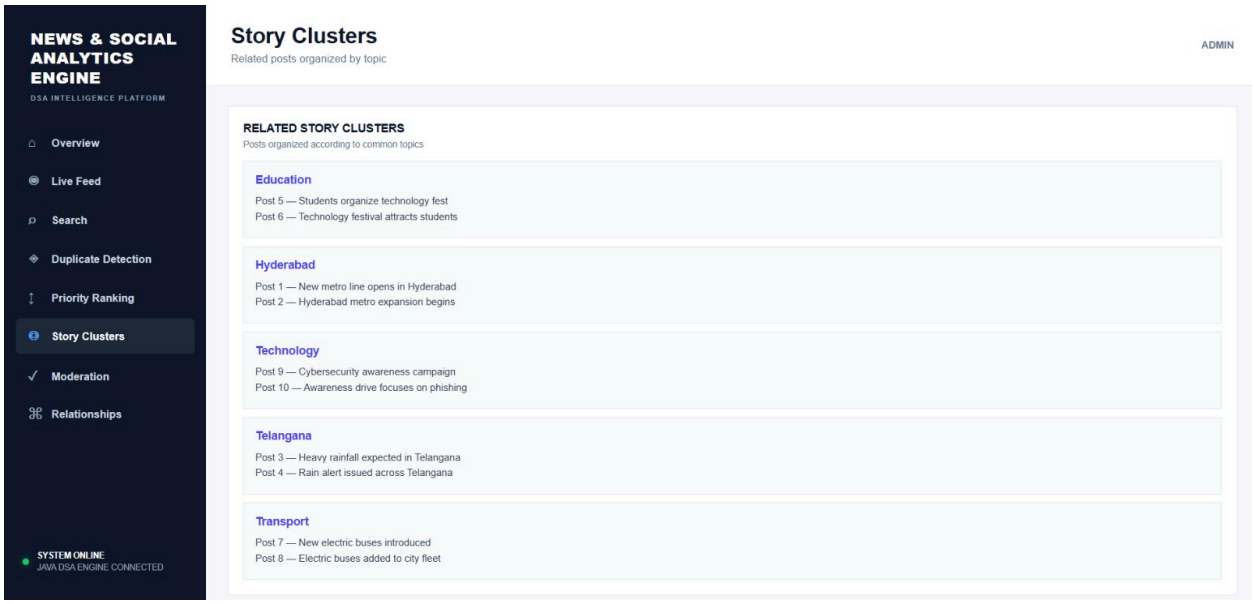
Priority Ranking

The Priority Ranking module arranges posts according to their priority scores using a max heap. Posts with higher priority scores appear first, helping users identify important content.



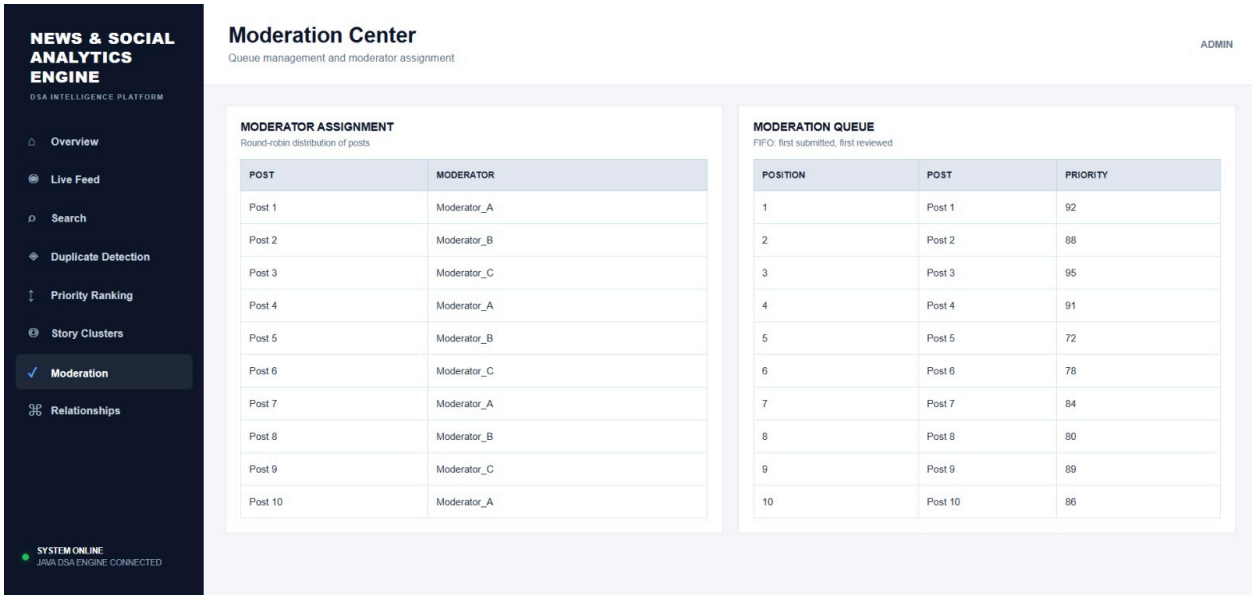
Story Clusters

The Story Clusters section groups related posts according to common topics. This makes it easier to view similar stories together, such as posts about Hyderabad or technology.



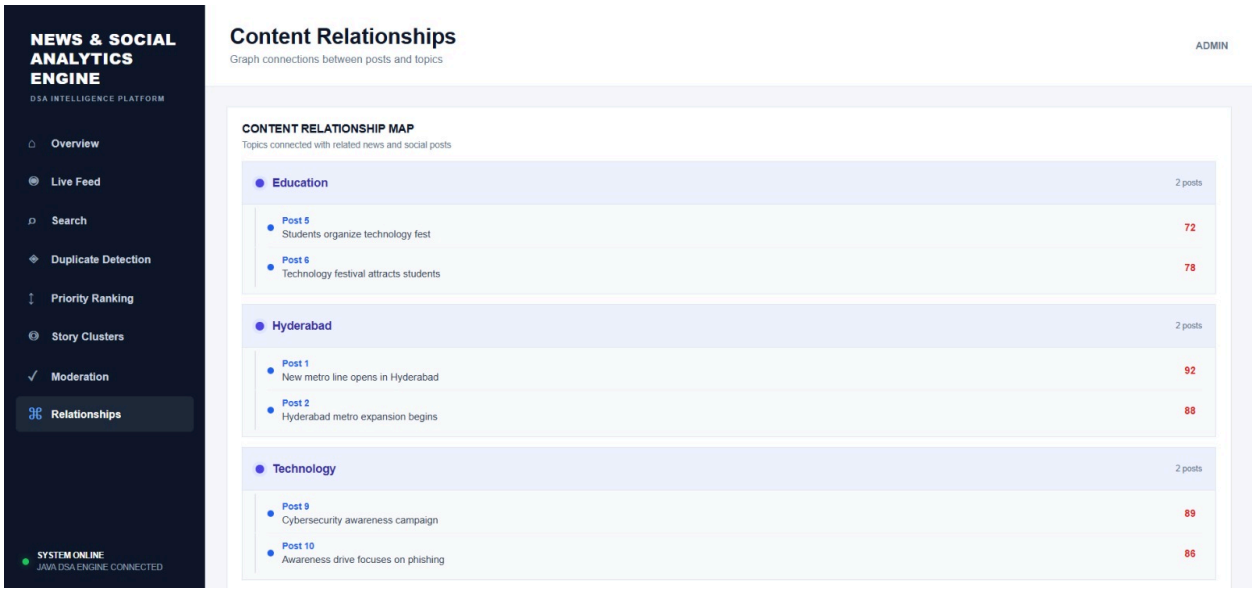
Moderation Center

The Moderation Center displays the moderation queue and the assignment of posts to moderators. It demonstrates how posts can be organized for review using queue-based processing.



Content Relationships

The Content Relationships section displays connections between topics and their related posts. It provides a structured view of how posts are associated with different topics in the dataset.



8. CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The News and Social Media Analytics Engine demonstrates how Data Structures and Algorithms can be applied to organize and analyze text-based content. The project combines a Java processing engine with a web-based dashboard and a Node.js/Express integration layer.

The implemented features include searching posts, duplicate detection, priority ranking, story grouping, moderation-related processing, and relationship visualization. The project provides a practical demonstration of data structures such as hash tables, heaps, queues, trees, and graphs in a single application.

The prototype achieves its main educational objective of demonstrating DSA concepts through a working interface using a sample dataset. Live data collection and production-scale processing remain possible areas for future development.

8.2 Limitations

The current application uses a sample CSV dataset rather than continuously collecting live posts from external platforms.

Duplicate detection and story grouping depend on the comparison and grouping rules implemented in the source code.

The prototype has not been evaluated at production scale with very large datasets.

The results depend on the quality and completeness of the input data.

Advanced semantic analysis and automated news verification are outside the scope of the current implementation unless specifically implemented in the source code.

8.3 Future Scope

The project can be improved in several ways:

Live Data Integration: Integrate authorized news APIs or social media APIs to retrieve current posts, subject to the relevant platform's terms and access requirements.

Advanced Duplicate Detection: Use text normalization and more advanced similarity methods to identify similar posts even when their wording differs.

Database Integration: Store posts and processing results in a database so that data can persist between application sessions.

Improved Search: Add filters for date, source, topic, and priority.

Enhanced Moderation: Add user authentication and role-based access for moderators.

Performance Evaluation: Test the system with larger datasets and record execution time and memory usage.

Testing and Reliability: Add automated tests, stronger error handling, and input validation for the Java modules and server routes.

9. REFERENCES

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